

Aaryan Kulkarni

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Electrical Engineering; GPA: 4.0

Atlanta, GA

May 2028

GT EE junior targeting digital design and hardware architecture roles; experienced in SystemVerilog RTL, RISC-V CPU design, real-time AI/CV systems, compound semiconductor research, and safety-critical embedded systems.

TECHNICAL SKILLS

Languages: Python, C++, SystemVerilog, Verilog

Hardware/RTL: RISC-V, FPGA architecture, digital logic, FSMs, MMIO, testbenches, VCD debug, Icarus Verilog, GTKWave

AI/CV: OpenCV, real-time video processing, object detection, alert logic, NumPy/SciPy, Matplotlib, Ollama

Embedded/Tools: Arduino Uno, HC-SR04, sensor integration, circuit debugging, Git/GitHub, VS Code, Arduino IDE

EXPERIENCE

Intern - Software Engineer

June 2026 – Present

Cloud Supply Chain Solutions

Alpharetta, GA

- Built a real-time AI driver safety monitoring prototype using Python, OpenCV, and live video to detect 4 fatigue-risk behaviors at approximately 20-30 FPS, including eye closure, yawning, head movement, and inattentiveness.
- Developed a dispatcher dashboard with live feeds, timestamped alerts, camera metadata, and clickable event snapshots.
- Implemented event-driven alert logic converting frame-level detections into actionable safety notifications.
- Integrated an Ollama-powered local AI agent to summarize fatigue-risk events and generate dispatcher-readable incident notes.

Undergraduate Research Assistant

Incoming Aug. 2026

Georgia Institute of Technology | Compound Semiconductor Products

Atlanta, GA

- Selected for undergraduate research supporting compound semiconductor product development, characterization, electrical testing, and lab documentation.
- Preparing to support device physics, materials characterization, data collection, technical reporting, and product-level engineering evaluation.

Co-Founder

Nov. 2023 – June 2024

Vizun

Atlanta, GA

- Co-founded an AR interior visualization startup enabling users to preview 20+ materials, layouts, and design concepts before purchase or renovation.
- Led product strategy translating 30+ customer discovery insights into visualization features, user stories, and early product requirements.
- Built financial models comparing 3 pricing strategies, startup costs, revenue assumptions, and customer acquisition channels.
- Secured \$2,700+ in seed funding and won 1st place at TiE Atlanta for innovation, market opportunity, and execution strategy.

PROJECTS

EvoFabric - Self-Adaptive RISC-V FPGA Compute Fabric | *SystemVerilog, Verification, GTKWave* June 2026

- Built a simulation-first RISC-V-style FPGA fabric with MMIO accelerator registers, runtime selection, safety fallback logic, and performance counters.
- Implemented Popcount, CRC32, FIR, and 2x2 matrix-multiply accelerator slots using shared start/busy/done/valid handshakes.
- Created 9 automated SystemVerilog testbenches for accelerator, controller, safety, and top-level MMIO verification.
- Added fault injection to verify checker mismatch, fallback activation, accelerator quarantine, and failed-call counter increments.

TinyRV32 - Modular RV32I Single-Cycle CPU Core | *SystemVerilog, RISC-V, Icarus Verilog, GTKWave* June 2026

- Designed a modular RV32I single-cycle CPU core in SystemVerilog with PC logic, instruction memory, ALU datapath, register file, and branch control.
- Implemented arithmetic, logic, load/store, branch, and jump instructions using reusable decode/control and datapath modules.
- Built module-level and top-level testbenches to verify PC updates, register writes, memory access, and branch/jump behavior through VCD analysis.

Scooter Distance Warning System | *Arduino, C++, HC-SR04*

Feb. 2026

- Built a scooter-mounted obstacle detection prototype using Arduino Uno, HC-SR04 ultrasonic sensing, and C++ to measure distance across a 2-400 cm range.
- Programmed 3 adaptive alert zones with buzzer frequency increasing as obstacle distance decreased; validated behavior across 25+ proximity tests.